
SMART STUDENT PRODUCTIVITY PLATFORM

RAGINI SINGH — OWN WORK DOCUMENTATION

Name	Ragini Singh
Roll No.	150096724023
Subject	Product Management & Software Development
Role	Solution Architect
Project	Smart Student Productivity Platform
Work Covered	Sprint 1 to Sprint 4

Introduction

This document explains my individual work in the Smart Student Productivity Platform from Sprint 1 to Sprint 4. My work started with understanding the project requirements and discussing the BRD with the PM and team. After that, I worked on the SRS, discussed the system architecture with the team, and finally moved to implementation work for the HOD Portal. My main implementation work was on the frontend, where I worked with the Tech Lead to build and integrate the HOD Portal into the existing project.

I have written the work sprint-wise so that it is clear what I worked on, how I worked with the team, and how the documentation and implementation were connected.

My Role in the Team

- Worked as the Solution Architect.
- Worked closely with the PM/BA for understanding and documenting requirements.
- Discussed technical flow and architecture with the team.
- Worked with the Tech Lead during frontend implementation and integration.
- Took responsibility for the HOD Portal frontend and its related authentication and routing work.

Sprint 1 — BRD Discussion and Requirement Understanding

During Sprint 1, my first focus was to understand what problem the platform was solving and what features were actually required. I discussed the requirements with the PM and the team instead of directly starting development. This helped me understand the users, their responsibilities and the overall academic workflow.

BRD Work

- Discussed the project requirements with the PM/BA and understood the business requirements.
- Helped in discussing and structuring the BRD based on the actual problem statement.
- Understood the main users of the system: Student, Faculty and HOD/Department Administrator, along with the overall administrator role.
- Discussed what each role should be able to see and manage.
- Focused especially on the HOD requirements such as department dashboard, faculty allocation, student monitoring, faculty monitoring, timetable, approvals and reports.
- Discussed the academic hierarchy and how department, program, semester, section, subject, faculty and students are connected.

My main learning in Sprint 1 was that development should start only after the requirements are clear. The BRD gave the team a common understanding of what we were building and why we were building it.

Sprint 2 — SRS Preparation and Functional Requirements

In Sprint 2, we moved from business requirements to software requirements. I worked on understanding how the BRD requirements could be converted into a proper SRS. I discussed the SRS structure and requirements with the team and worked on documenting the system in a more technical and implementation-friendly way.

SRS Work

- Worked on the overall SRS structure and requirement organization.
- Documented the system purpose, scope and objectives.
- Worked on functional requirements and role-wise access.
- Discussed the authentication and authorization flow.
- Worked on Role-Based Access Control (RBAC) and academic-scope restrictions.
- Connected the HOD requirements from the BRD with their corresponding software functionality in the SRS.
- Reviewed the HOD modules so that the requirements were properly represented before implementation.

The SRS helped us move from the question 'What does the system need?' to 'How should the software behave?' This became the base for the architecture and development work in the next sprint.

Sprint 3 — Architecture Discussion and System Flow

In Sprint 3, my focus shifted towards the technical side of the project. I discussed the overall architecture with the team and worked on understanding how the different portals, authentication, database, backend services and external services would work together.

Architecture Work

- Discussed the high-level architecture with the team.
- Mapped the Student, Faculty and HOD users to their respective portals.
- Worked with the frontend flow using HTML5, CSS3 and JavaScript.
- Discussed Firebase Authentication for login and role identification.
- Mapped Firestore and Firebase services to data and backend operations.
- Discussed Cloud Functions for business logic and service integration.
- Considered Firebase Cloud Messaging for notifications.
- Discussed external integrations such as Google Meet, Clouinary and the AI service.
- Focused on how role-based routing and protected pages should work.

The architecture discussions helped me understand how the BRD and SRS would actually become a working software system. I also used these discussions while planning the HOD Portal implementation.

Sprint 4 — HOD Portal UI/UX and Frontend Development

Sprint 4 was the main implementation phase for my work. I took responsibility for building the HOD/Department Administrator Portal on the frontend. I worked with the Tech Lead so that my changes followed the existing project structure and did not unnecessarily affect the Student and Faculty portals.

HOD Portal Work

- Created the HOD Portal structure and separate HOD pages.
- Built the HOD Dashboard with department-level information and monitoring sections.
- Worked on Faculty Allocation functionality.
- Created Student Monitoring and Faculty Monitoring pages.
- Created Timetable Management page.
- Created Academic Approvals page.
- Created Reports page.
- Created Settings and Profile pages.
- Worked on HOD-specific CSS and frontend styling.
- Created mock HOD data and frontend service logic where required for the portal.
- Implemented HOD authentication guard and role-based routing.
- Integrated the HOD role into the existing login flow while keeping the existing Student and Faculty flows.
- Worked on matching the HOD Portal UI with the existing Student Portal design so the project looks consistent.

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- Tested the HOD pages and navigation during implementation and fixed frontend issues found during the work.
 - Worked with the Tech Lead during integration and Git/GitHub workflow for the frontend repository.

This sprint was important for me because I moved from documentation and architecture into actual implementation. I could directly see how the requirements and architecture discussed in the earlier sprints were converted into working screens and flows.

Overall Work from Sprint 1 to Sprint 4

Overall, my work followed a complete flow: first understanding the requirements with the PM, then helping structure the BRD, preparing the SRS, discussing the architecture with the team, and finally implementing the HOD Portal frontend with the Tech Lead. This gave me experience in both documentation and practical software development.

- Sprint 1: BRD discussion, requirement understanding and HOD requirement analysis.
- Sprint 2: SRS preparation, functional requirements, authentication and access-control understanding.
- Sprint 3: Architecture discussion, system flow and technology/service mapping.
- Sprint 4: HOD Portal UI/UX, frontend development, authentication/routing integration and Git/GitHub work.

My Key Contribution

My main individual contribution was the HOD/Department Administrator Portal. The portal was created as a separate role-based area of the Smart Student Productivity Platform. It covers the department-level activities required from the BRD and SRS, including dashboard monitoring, faculty allocation, student and faculty monitoring, timetable management, approvals and reports.

I also contributed before development started. The BRD and SRS work helped me understand the project at a system level instead of only working on screens. The architecture discussions then helped me connect the requirements with authentication, role-based access, Firebase services and the frontend structure.

Team Collaboration

- PM/BA: Discussed requirements and BRD content.
- Project Team: Discussed SRS requirements and system architecture.
- Tech Lead: Coordinated frontend implementation, project structure, integration and Git/GitHub workflow.
- My Responsibility: Solution-architecture contribution and HOD Portal frontend implementation.

Conclusion

From Sprint 1 to Sprint 4, my work gradually moved from requirement understanding to documentation, architecture and implementation. I did not work only on one isolated screen; I worked through the project flow and then implemented the HOD Portal based on the requirements discussed earlier.

The biggest part of my practical contribution was the HOD Portal frontend. At the same time, the BRD, SRS and architecture work helped me understand why each feature was required and how it should fit into the overall platform. Working with the PM, project team and Tech Lead also gave me practical experience of collaborative software development.

Ragini Singh

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